

● robot / microrobot ● patient hand

a

Simulation environment

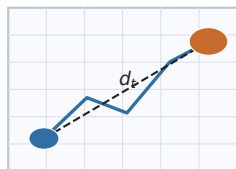


table-top pursuit task

CMD-DR virtual patient

intent: SPC / learned hand

motor layer: delay, noise, LPF, accel. limit

ZPD reward + boundary

O_t, r_t

a_t

same task

b

Policy and league learning

adaptive robot policy

scalar MLP

history GRU

fusion

action head

aux. future-dynamics head

hand pool
 $H_1...H_k$ + SPC

PFSP sampler

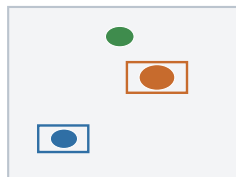
PPO update

next robot

trained policy

c

Vision and state estimation



camera frame / detection overlay

async vision thread

YOLO microrobot

MediaPipe hand

state buffer +
dead reckoning

image feed

real O_t

d

Real-world rehabilitation platform



annotated real setup to be inserted

20 Hz
control thread

UR10 RTDE

workspace
inset

magnetic
actuation